

HR Attrition Analysis — Executive Summary

Business Question

What factors are associated with employee resignation, and which departments are at risk

Dataset

IBM HR Analytics dataset (1470 employees).

Key Metrics (Definitions)

Attrition Rate = (# Attrition = Yes) / (Total Employees)

At-risk group = attrition rate above overall attrition rate

Key Findings (from analysis)

- 1) Highest-risk department: Sales (~20% attrition rate).
- 2) OverTime is a strong risk factor: OverTime=Yes 30.53% vs OverTime=No 10.44%.
- 3) Highest-risk job role: Sales Representative 39.76% attrition.
- 4) Early-tenure employees are most at risk: Tenure 0–1 years 34.88% attrition.

Recommendations

- 1) Reduce sustained overtime in high-attrition teams (monitor weekly OT hours + staffing).
- 2) Strengthen onboarding + manager check-ins for first-year employees (0–1 year tenure).
- 3) Review Sales Representative role: workload, incentives, career path, manager support.
- 4) Track attrition monthly by Department × Tenure band to catch risk early (use the heatmap).